

TCFI Graph Generation

Georg Schildbach

March 2026

1 Table

TC	Triggered FI
(TC001) traffic congestion	(FI001) no capability to form a rescue alley
(TC001) traffic congestion	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC002) obstacle avoidance scenario	(FI003) no emergency maneuvers w/o lane change / shoulder
(TC002) obstacle avoidance scenario	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC003) blocking obstacle shortly beyond prediction horizon N	(FI004) prediction horizon too short to stop fully
(TC004a) end of two merging lanes	(FI005a) cannot ensure sufficient progress
(TC004a) end of two merging lanes	(FI005b) cannot avoid deadlocks
(TC004a) end of two merging lanes	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC004b) special situations during general lane changes	(FI005a) cannot ensure sufficient progress
(TC004b) special situations during general lane changes	(FI005b) cannot avoid deadlocks
(TC004b) special situations during general lane changes	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC005) leading target vehicle driving in reverse	(FI006) cannot generate feasible collision-free trajectory
(TC006) wrong way driver	(FI006) cannot generate feasible collision-free trajectory
(TC006) wrong way driver	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC007a) water (hydroplaning)	(FI006) cannot generate feasible collision-free trajectory
(TC007b) snow/ice	(FI006) cannot generate feasible collision-free trajectory
(TC007c) oil spillage	(FI006) cannot generate feasible collision-free trajectory
(TC007a) water (hydroplaning)	(FI007a) due to linear tire model in LDBM
(TC007b) snow/ice	(FI007a) due to linear tire model in LDBM
(TC007c) oil spillage	(FI007a) due to linear tire model in LDBM
(TC008a) gravel or stones	(FI007b) due to irregular road surface
(TC008b) potholes	(FI007b) due to irregular road surface
(TC008c) road bumps	(FI007b) due to irregular road surface
(TC009a) bridge deck	(FI008) receding-horizon + tracking may oscillate/unstable

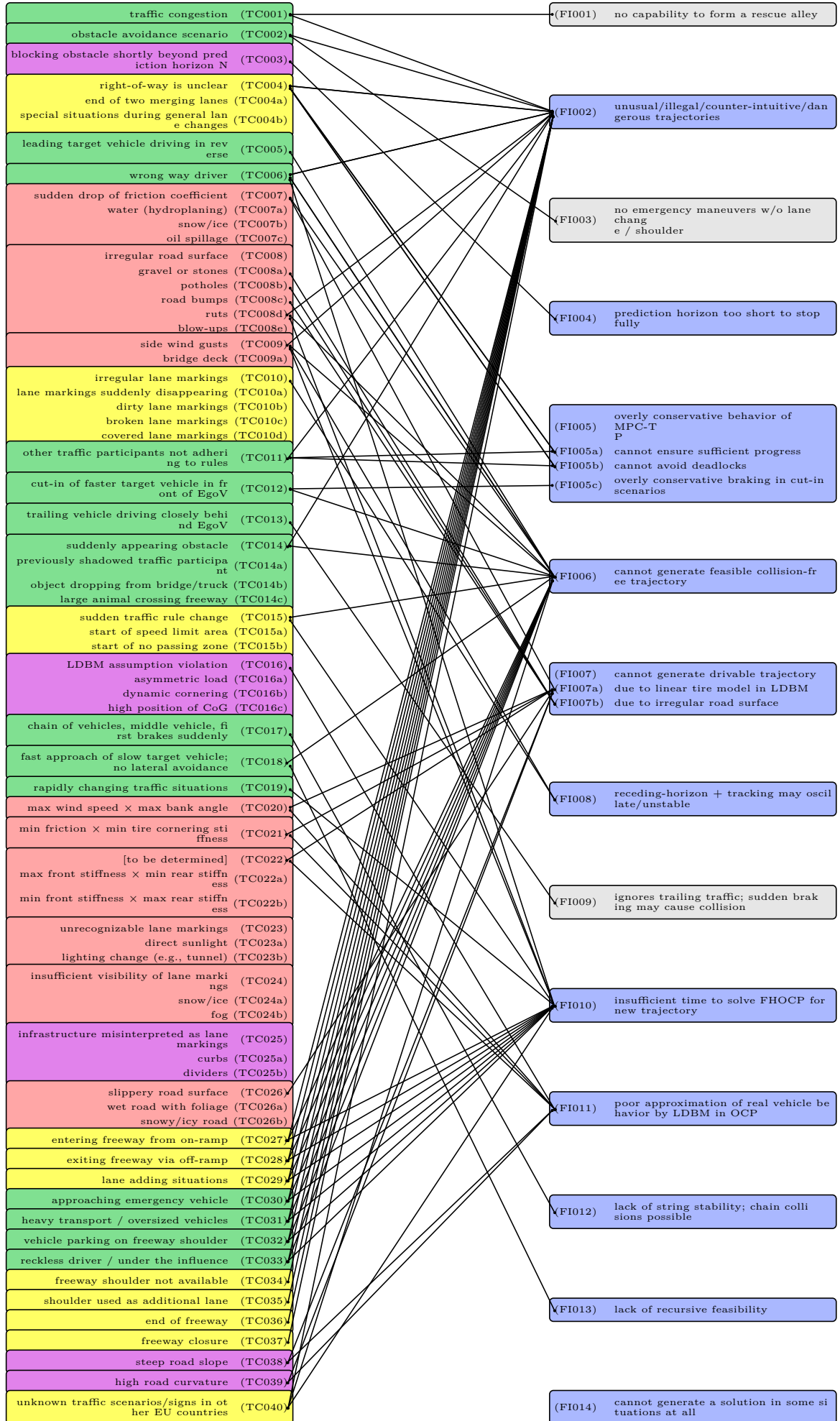
(TC010a) lane markings suddenly disappearing	(FI008) receding-horizon + tracking may oscillate/unstable
(TC010b) dirty lane markings	(FI008) receding-horizon + tracking may oscillate/unstable
(TC010c) broken lane markings	(FI008) receding-horizon + tracking may oscillate/unstable
(TC010d) covered lane markings	(FI008) receding-horizon + tracking may oscillate/unstable
(TC004) right-of-way is unclear	(FI005a) cannot ensure sufficient progress
(TC004) right-of-way is unclear	(FI005b) cannot avoid deadlocks
(TC004) right-of-way is unclear	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC011) other traffic participants not adhering to rules	(FI005a) cannot ensure sufficient progress
(TC011) other traffic participants not adhering to rules	(FI005b) cannot avoid deadlocks
(TC011) other traffic participants not adhering to rules	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC012) cut-in of faster target vehicle in front of EgoV	(FI005c) overly conservative braking in cut-in scenarios
(TC012) cut-in of faster target vehicle in front of EgoV	(FI006) cannot generate feasible collision-free trajectory
(TC013) trailing vehicle driving closely behind EgoV	(FI009) ignores trailing traffic; sudden braking may cause collision
(TC014a) previously shadowed traffic participant	(FI006) cannot generate feasible collision-free trajectory
(TC014b) object dropping from bridge/truck	(FI006) cannot generate feasible collision-free trajectory
(TC014c) large animal crossing freeway	(FI006) cannot generate feasible collision-free trajectory
(TC014a) previously shadowed traffic participant	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC014b) object dropping from bridge/truck	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC014c) large animal crossing freeway	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC015a) start of speed limit area	(FI006) cannot generate feasible collision-free trajectory
(TC015b) start of no passing zone	(FI006) cannot generate feasible collision-free trajectory
(TC015a) start of speed limit area	(FI010) insufficient time to solve FHOCF for new trajectory
(TC015b) start of no passing zone	(FI010) insufficient time to solve FHOCF for new trajectory
(TC016a) asymmetric load	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC016b) dynamic cornering	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC016c) high position of CoG	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC017) chain of vehicles, middle vehicle, first brakes suddenly	(FI012) lack of string stability; chain collisions possible

(TC018) fast approach of slow target vehicle; no lateral avoidance	(FI006) cannot generate feasible collision-free trajectory
(TC018) fast approach of slow target vehicle; no lateral avoidance	(FI013) lack of recursive feasibility
(TC019) rapidly changing traffic situations	(FI010) insufficient time to solve FHOCPP for new trajectory
(TC040) unknown traffic scenarios/signs in other EU countries	(FI014) cannot generate a solution in some situations at all
(TC020) max wind speed $\times$ max bank angle	(FI007a) due to linear tire model in LDBM
(TC021) min friction $\times$ min tire cornering stiffness	(FI007a) due to linear tire model in LDBM
(TC022a) max front stiffness $\times$ min rear stiffness	(FI007a) due to linear tire model in LDBM
(TC022b) min front stiffness $\times$ max rear stiffness	(FI007a) due to linear tire model in LDBM
(TC020) max wind speed $\times$ max bank angle	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC021) min friction $\times$ min tire cornering stiffness	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC022a) max front stiffness $\times$ min rear stiffness	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC022b) min front stiffness $\times$ max rear stiffness	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC010a) lane markings suddenly disappearing	(FI014) cannot generate a solution in some situations at all
(TC010b) dirty lane markings	(FI014) cannot generate a solution in some situations at all
(TC010c) broken lane markings	(FI014) cannot generate a solution in some situations at all
(TC010d) covered lane markings	(FI014) cannot generate a solution in some situations at all
(TC023a) direct sunlight	(FI014) cannot generate a solution in some situations at all
(TC023b) lighting change (e.g., tunnel)	(FI014) cannot generate a solution in some situations at all
(TC024a) snow/ice	(FI014) cannot generate a solution in some situations at all
(TC024b) fog	(FI014) cannot generate a solution in some situations at all
(TC025a) curbs	(FI014) cannot generate a solution in some situations at all
(TC025b) dividers	(FI014) cannot generate a solution in some situations at all
(TC008b) potholes	(FI014) cannot generate a solution in some situations at all
(TC008c) road bumps	(FI014) cannot generate a solution in some situations at all
(TC026a) wet road with foliage	(FI007a) due to linear tire model in LDBM
(TC026b) snowy/icy road	(FI007a) due to linear tire model in LDBM
(TC027) entering freeway from on-ramp	(FI010) insufficient time to solve FHOCPP for new trajectory
(TC028) exiting freeway via off-ramp	(FI010) insufficient time to solve FHOCPP for new trajectory
(TC029) lane adding situations	(FI010) insufficient time to solve FHOCPP for new trajectory

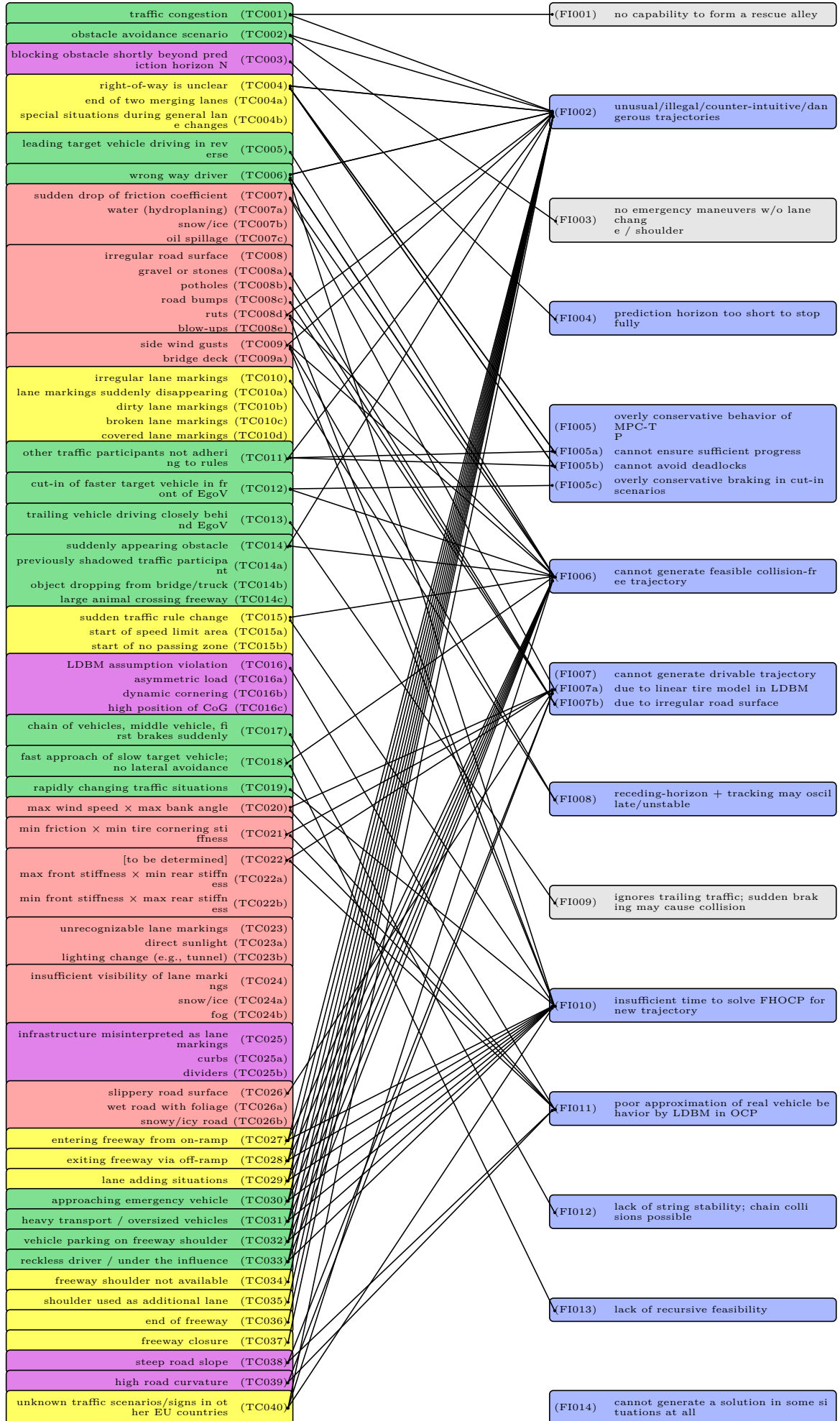
(TC027) entering freeway from on-ramp	(FI006) cannot generate feasible collision-free trajectory
(TC028) exiting freeway via off-ramp	(FI006) cannot generate feasible collision-free trajectory
(TC029) lane adding situations	(FI006) cannot generate feasible collision-free trajectory
(TC027) entering freeway from on-ramp	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC028) exiting freeway via off-ramp	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC029) lane adding situations	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC006) wrong way driver	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC030) approaching emergency vehicle	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC031) heavy transport / oversized vehicles	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC032) vehicle parking on freeway shoulder	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC033) reckless driver / under the influence	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC006) wrong way driver	(FI010) insufficient time to solve FHOC for new trajectory
(TC030) approaching emergency vehicle	(FI010) insufficient time to solve FHOC for new trajectory
(TC031) heavy transport / oversized vehicles	(FI010) insufficient time to solve FHOC for new trajectory
(TC032) vehicle parking on freeway shoulder	(FI010) insufficient time to solve FHOC for new trajectory
(TC033) reckless driver / under the influence	(FI010) insufficient time to solve FHOC for new trajectory
(TC006) wrong way driver	(FI006) cannot generate feasible collision-free trajectory
(TC030) approaching emergency vehicle	(FI006) cannot generate feasible collision-free trajectory
(TC031) heavy transport / oversized vehicles	(FI006) cannot generate feasible collision-free trajectory
(TC032) vehicle parking on freeway shoulder	(FI006) cannot generate feasible collision-free trajectory
(TC033) reckless driver / under the influence	(FI006) cannot generate feasible collision-free trajectory
(TC008d) ruts	(FI010) insufficient time to solve FHOC for new trajectory
(TC040) unknown traffic scenarios/signs in other EU countries	(FI010) insufficient time to solve FHOC for new trajectory
(TC008d) ruts	(FI006) cannot generate feasible collision-free trajectory
(TC040) unknown traffic scenarios/signs in other EU countries	(FI006) cannot generate feasible collision-free trajectory
(TC008d) ruts	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories

(TC040) unknown traffic scenarios/signs in other EU countries	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC034) freeway shoulder not available	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC035) shoulder used as additional lane	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC036) end of freeway	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC037) freeway closure	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC038) steep road slope	(FI007a) due to linear tire model in LDBM
(TC039) high road curvature	(FI007a) due to linear tire model in LDBM
(TC038) steep road slope	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC039) high road curvature	(FI011) poor approximation of real vehicle behavior by LDBM in OCP
(TC009) side wind gusts	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC009) side wind gusts	(FI006) cannot generate feasible collision-free trajectory
(TC009) side wind gusts	(FI010) insufficient time to solve FHOCPP for new trajectory
(TC040) unknown traffic scenarios/signs in other EU countries	(FI002) unusual/illegal/counter-intuitive/dangerous trajectories
(TC040) unknown traffic scenarios/signs in other EU countries	(FI006) cannot generate feasible collision-free trajectory
(TC040) unknown traffic scenarios/signs in other EU countries	(FI010) insufficient time to solve FHOCPP for new trajectory

2 Original Graph



3 Numerically Ordered Graph



4 Optimized Graph

